AI Capital versus Regular Capital: Real Interest Rates, Wages and the Labor Share after a Jump or an S-Curve

Research note (draft)

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Abstract

This note compares the response of the real interest rate, the real wage and the labor share to technical progress in AI capital and in regular capital. Regular capital and labor form a Cobb-Douglas bundle, and AI capital substitutes for that bundle ($\varepsilon = 1.5$). Progress is either disembodied (in the production function) or embodied (a lower price of new capital in the accumulation equation). Both capital stocks face the same moderate adjustment costs. Two experiments are considered, a surprise permanent jump and an S-curve centered on year 10 and known at year 0. Elasticities are the same in every experiment, and every shock raises long-run output by the same 10 percent. In the long run, which is in closed form, AI progress raises the wage by 7.4 percent and lowers the labor share by 1.8 points. Regular-capital progress raises the wage by the full 10 percent and leaves the labor share unchanged. In the short run, the interest rate rises most for embodied AI progress after a jump, and most for disembodied regular-capital progress along an S-curve. Wage gains arrive fastest under disembodied regular-capital progress and slowest under embodied progress of either kind.

1 Model

Notation follows Kiley (2000). Production is

$$X = (\Gamma_K K)^a (ZN)^{1-a}, \qquad Y = \left[\omega^{1/\varepsilon} X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} (\Gamma_A A)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon = 1.5,$$

so AI capital A substitutes for both labor and regular capital K. Capital is measured in efficiency units and accumulates as $K_j' = (1-d_j)K_j + I_j/P_j$, where P_j is the output price of a new efficiency unit. Adjustment costs, following Kiley (2004), are

$$P_j \tfrac{c_j}{2} (i_j - x_j)^2 K_j, \qquad i_j = \frac{I_j}{P_j K_j}, \qquad c_j = \frac{1}{x_j}.$$

They vanish at the balanced-growth investment rate $x_j = g + d_j$, and they give both stocks the same unit elasticity of investment with respect to Q_j/P_j (the paper's "moderate" case). *Disembodied* progress raises Γ_j, which improves the installed stock. *Embodied* progress lowers P_j, which improves only new investment. Utility is logarithmic and hours are fixed.

Calibration. $a = 0.282$, $d_K = 0.066$, $d_A = 0.20$, trend growth $g = 1.5$ percent, initial $r = 7.5$ percent, initial AI income share 3 percent, initial labor share 69.6 percent.

Proposition 1 (Balanced growth path)**.** *With constant technology levels, $Q_j = P_j$, $1+r = (1+g)/B$ and $u_j = r + d_j$. Let $\lambda_j = P_j u_j/\Gamma_j$ be the cost of an effective unit of capital j. Then*

$$s_A = (1-\omega)\lambda_A^{1-\varepsilon}, \qquad \textit{labor share} = (1-a)(1-s_A), \qquad P_X = \Big(\frac{1-s_A}{\omega}\Big)^{\frac{1}{1-\varepsilon}},$$
$$\ln\frac{W}{Z} = \frac{\ln P_X}{1-a} - \frac{a}{1-a}\ln\lambda_K + \textit{const}, \qquad \ln\frac{Y}{Z} = \ln\frac{W}{Z} - \ln(1-s_A) + \textit{const}.$$

Proposition 2 (Long run, same output gain γ_Y)**.** *Disembodied and embodied progress have the same long-run effects ($\ln\Gamma_j = -\ln P_j$), and the interest rate returns to its initial level.*

1. Regular capital*:* $\ln(\Gamma_K/P_K) = \frac{1-a}{a}\gamma_Y$*, $\Delta\ln W = \gamma_Y$, and the labor share is unchanged.*
2. AI capital*:* $\Delta\ln W = \dfrac{\gamma_Y}{1+(1-a)(\varepsilon-1)}$ *and* $\Delta\ln(\textit{labor share}) = -\dfrac{(1-a)(\varepsilon-1)}{1+(1-a)(\varepsilon-1)}\gamma_Y$.

Proof. By Proposition 1, $\ln(Y/Z) = \big[\frac{1}{(1-a)(1-\varepsilon)} - 1\big]\ln(1-s_A) - \frac{a}{1-a}\ln\lambda_K + \text{const}$, and $\ln W = \ln Y + \ln(1-s_A)$. Regular-capital progress moves only λ_K, and AI progress moves only s_A. □

For $\gamma_Y = 10$ percent, regular-capital technology rises 29 percent, against 3.4-fold for AI, because AI starts small. AI's income share rises from 3.0 to 5.5 percent. In the long run, AI progress raises the wage 7.4 percent and lowers the labor share from 69.6 to 67.8 percent. Regular-capital progress raises the wage 10 percent and leaves the labor share unchanged.

2 Results

Both experiments start on the initial balanced growth path. In the *jump*, technology rises permanently and unexpectedly at $t = 0$. In the *S-curve*, a logistic rise centered on year 10 is announced at $t = 0$. Each is solved as a nonlinear perfect-foresight problem. Wages are log deviations from the pre-shock path, and the labor share is relative to gross output.

Table 1: Summary. r_1 is the real rate between years 0 and 1. All shocks raise long-run output by 10%.

		real interest rate (pct.)			real wage, $\ln W$ (pct.)				labor share (pct.)	
		r_1	peak	year	t=0	t=10	t=20	long run	t=10	long run
Jump	Regular, disemb.	7.69	7.69	0	7.0	8.4	9.1	10.0	69.7	69.6
	Regular, emb.	8.14	8.14	0	0.0	4.4	6.9	10.0	69.7	69.6
	AI, disemb.	8.05	8.05	0	3.0	5.5	6.4	7.4	68.0	67.8
	AI, emb.	8.73	8.73	0	0.0	4.5	6.0	7.4	68.2	67.8
S-curve	Regular, disemb.	7.57	8.34	10	0.0	3.7	8.1	10.0	69.7	69.6
	Regular, emb.	7.50	7.90	16	0.0	0.6	4.0	10.0	69.7	69.6
	AI, disemb.	7.53	8.15	11	0.0	2.1	5.4	7.4	68.9	67.8
	AI, emb.	7.50	7.99	15	0.0	0.8	4.1	7.4	69.4	67.8

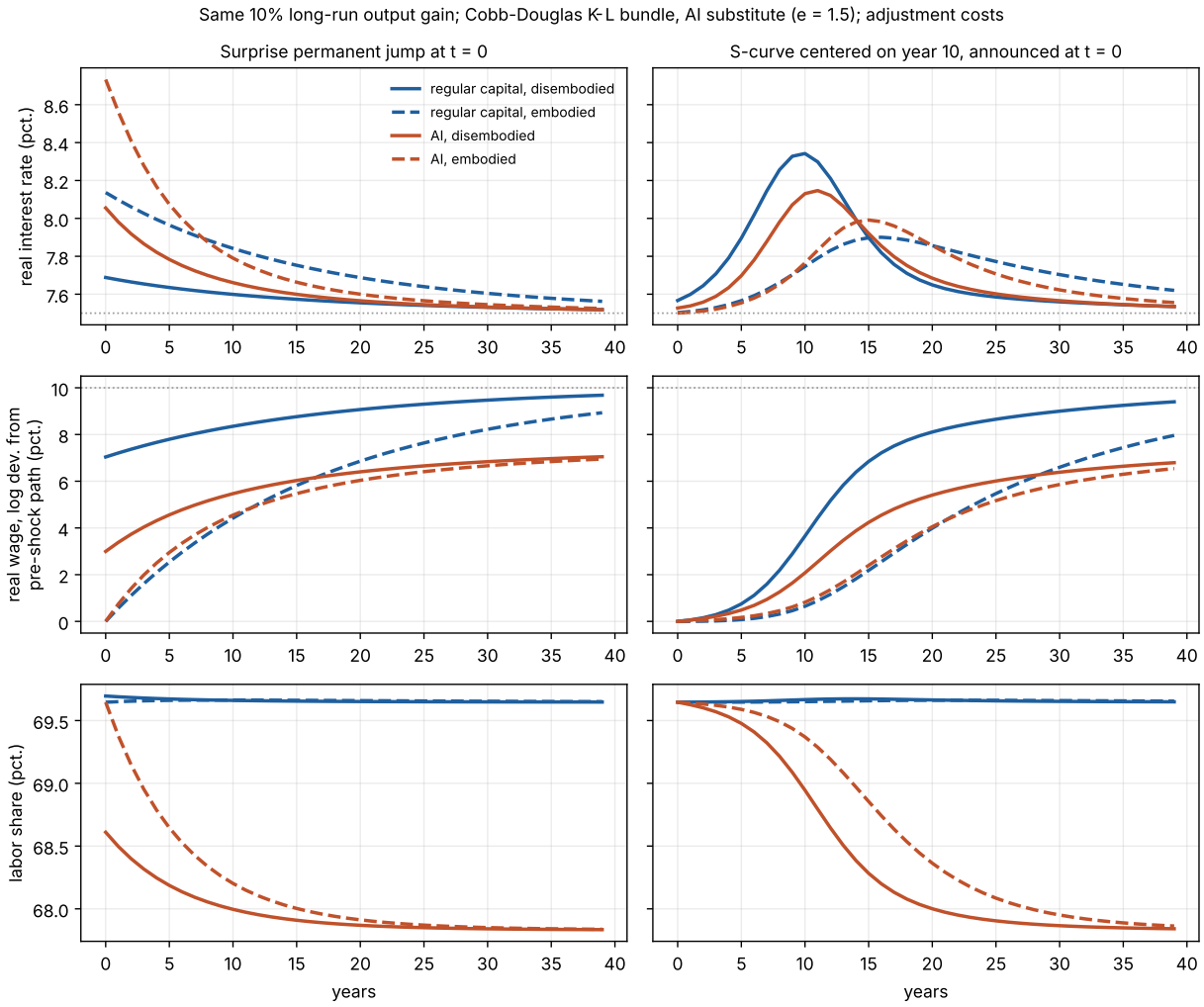


Figure 1: Same 10% long-run output gain in every experiment. Dotted lines: initial r and long-run output gain.

Real interest rate

- *Jump.* The rate jumps at once and then declines. On impact the ranking is AI embodied (8.7 percent), regular embodied (8.1), AI disembodied (8.05) and regular disembodied (7.7). Embodied progress raises r more because the output gain requires an investment boom first. AI raises r more than regular capital because a larger share of its long-run gain must be accumulated. Disembodied AI delivers 3.0 of its 7.4 percent wage gain on impact, against 7.0 of 10 percent for regular capital. Over 40 years, the excess return adds up to about 9 percentage points in both embodied cases, against 4.7 for disembodied AI and 2.7 for disembodied regular capital.

- *S-curve.* The rate barely moves at announcement. Under disembodied progress it rises ahead of the innovation and peaks with it (year 10–11): 8.3 percent for regular capital and 8.15 for AI. Under embodied progress it peaks later (years 15–16) and lower, at 7.9 for regular capital and 8.0 for AI, but the response lasts longer. The 40-year cumulative effects are similar,

7.4–9.4 percentage points.

Real wage

- *Long run*: 10.0 percent for regular capital and 7.4 percent for AI (Proposition 2).
- *Jump.* Disembodied progress raises the wage on impact, by 7.0 percent for regular capital and 3.0 for AI. Embodied progress starts from zero. By year 10 the wage is up 8.4 (regular, disembodied), 5.5 (AI, disembodied), 4.5 (AI, embodied) and 4.4 percent (regular, embodied).
- *S-curve.* The wage follows the innovation, and much more slowly when progress is embodied. At year 10 it is up 3.7 percent (regular, disembodied), 2.1 (AI, disembodied), 0.8 (AI, embodied) and 0.6 (regular, embodied). At year 20 the figures are 8.1, 5.4, 4.1 and 4.0.

Labor share

- *Regular capital*: essentially unchanged throughout, within 0.05 points.
- *AI*: falls 1.8 points to 67.8 percent. After a disembodied jump, about 60 percent of the fall happens on impact. After an embodied jump it happens as AI capital is installed (68.2 percent at year 10). Along the S-curve it tracks the innovation, at 68.9 percent (disembodied) or 69.4 percent (embodied) in year 10.

3 Summary

With the same elasticities and the same long-run output gain:

1. AI progress delivers about three-quarters of the wage gain of regular-capital progress and lowers the labor share. Regular-capital progress leaves the labor share unchanged. These long-run results are in closed form and do not depend on whether progress is embodied or disembodied.
2. The interest rate rises temporarily in every case. A surprise jump raises it most when progress is embodied and in AI. An anticipated S-curve raises it most, and soonest, when progress is disembodied. Embodied progress produces a later, smaller but more persistent rise.
3. Embodied progress delays wage gains by roughly five to ten years compared with disembodied progress. Embodied AI and embodied regular-capital progress produce similar wage paths in the first decade.

A Verification

`capital_cases.py` produces all results. It checks that the closed-form balanced growth path satisfies the equilibrium conditions, that every experiment converges to the 10 percent output gain, and that long-run wages match Proposition 2. Newton residuals are below 10^{-10}.

Check	max abs. error
closed-form BGP satisfies equilibrium	1.3e-15
jump K dis: Newton residual	2.7e-12
jump K dis: long-run ln Y gain vs 10jump K emb: Newton residual	8.5e-13
jump K emb: long-run ln Y gain vs 10jump A dis: Newton residual	3.0e-15
jump A dis: long-run ln Y gain vs 10jump A emb: Newton residual	3.8e-13
jump A emb: long-run ln Y gain vs 10scurve K dis: Newton residual	3.9e-12
scurve K dis: long-run ln Y gain vs 10scurve K emb: Newton residual	2.9e-13
scurve K emb: long-run ln Y gain vs 10scurve A dis: Newton residual	3.4e-11
scurve A dis: long-run ln Y gain vs 10scurve A emb: Newton residual	2.1e-15
scurve A emb: long-run ln Y gain vs 10jump K dis: long-run ln W vs closed form	3.4e-08
jump K emb: long-run ln W vs closed form	1.1e-07
scurve K dis: long-run ln W vs closed form	6.3e-08
scurve K emb: long-run ln W vs closed form	2.1e-07
jump A dis: long-run ln W vs closed form	3.4e-08
jump A emb: long-run ln W vs closed form	4.5e-08
scurve A dis: long-run ln W vs closed form	6.1e-08
scurve A emb: long-run ln W vs closed form	8.5e-08

References

Kiley, M. T. (2000). "Stock Prices and Fundamentals in a Production Economy." FEDS Working Paper 2000-05, Federal Reserve Board.

Kiley, M. T. (2004). "Stock Prices and Fundamentals: A Macroeconomic Perspective." *Journal of Business* 77(4): 909–936.